

SIDDHANT K. SANCHETI

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SKILLS

Languages: Python, Java, Ruby, Bash, TypeScript, JavaScript, SQL, HTML5, CSS3, Shell.

Tools & Infrastructure: Linux, Kubernetes, Docker, CI/CD (Jenkins, GitHub Actions), AWS (Lambda, EC2, S3, RDS, ECS, VPC, EBS, Route53), Observability (CloudWatch, CloudTrail, Grafana), GCP, Asynchronous/Concurrent Programming, RESTful APIs, RPC, Git.

Frameworks: AWS CDK, React, Node.js, Express.js, Flask, Django, Apache Spark, Bootstrap, Springboot, Kafka.

Databases: AWS RDS, DynamoDB, PostgreSQL, MongoDB, Firebase.

AI/ML Stack: TensorFlow, PyTorch, JAX, LangChain, VertexAI, FAISS, Computer Vision, OpenCV, Pandas, NumPy, TensorRT, Keras, MCP, Sagemaker, LLMs, RAG, Natural Language Processing.

WORK EXPERIENCE

Software Development Engineer, Amazon Web Services, Seattle, WA

May 2025 - Present

RDS SQL to IAM auth migration (Primary Skill: CDK(IaC), Java, CloudWatch, Lambda, TypeScript, RDS, SQL, Ruby)

- Led zero-downtime migration of **Aurora PostgreSQL** to IAM auth across **110+ Lambda** services and **CLI** tools in **17** regions, supporting **150-190** connections/sec, achieving **100%** IAM adoption with **0** password fallback, and resolving key production issues to stabilize high-throughput multi-region workloads.

On-Call Operations (Primary Skill: Problem Solving, Triaging, RCA, Bash, Change Management, Production Support)

- Owned on-call operations for production AWS services, resolving **110+** tickets by triaging incidents, partnering with customers, shipping CR-based fixes, and authoring release management with rollback plans for controlled lifecycle changes, improving service reliability and preventing rollout-related failures.

AI Agent Orchestration for code automation (Primary Skill: Python, CDK, Go, DAG, PostgreSQL)

- Co-designed and built a **Python multi-agent orchestration harness** that parsed task tables into dependency DAGs and scheduled **5-10** asynchronous **parallel agent workflows** with persistent session state, resumable execution, and live observability (logging/metrics), cutting 10-task delivery cycles across a **195+** package ecosystem from **5-7 days** to **1-2 hours**.

Fleet Capacity Offset Management System (Primary Skill: Java, DynamoDB, Coral RPC, CLI Tooling, Shell)

- Built a self-service fleet capacity management system using **Java, DynamoDB, RPC APIs**, and safe-by-default CLIs, reducing scaling workflows from **30 mins** to **2-4 mins** across **5,000+ EC2** hosts while preventing conflicting updates and error-prone manual production changes.

Software Engineer - ML/AI, GrantAide, San Jose, USA

Sept 2024 - Jan 2025

AI-Driven Grant Writing Platform (Primary Skill: Python, React, TypeScript, GPT-4, Flask, AWS, Vertex AI, LangChain, FAISS)

- Scaled and built an AI-driven grant writing platform using **React, GPT-4, Flask, AWS (S3, Amplify, EC2, Elastic Beanstalk)** and **Stripe APIs** improving platform responsiveness **5x** and strengthening payments, subscriptions, and overall user experience.
- Engineered responsive frontend and **RAG** workflows with **Material-UI, Firebase, Vertex AI, LangChain**, and **FAISS**, reducing frontend load times by **35%**, cutting query latency from **200ms** to **50ms**, improving response accuracy by **80%**, and increasing grant application success rates by **85%**.

Software Developer - Full Stack, Forks Technologies, Jaipur, India

Mar 2021 - Jun 2022

Campus Portal Modernization (Primary Skill: React, Node.js, Express.js, GraphQL, AWS, Jenkins, Docker)

- Architected an AI-enabled Campus Portal using **React, Node.js, Express.js, AWS Lambda, DynamoDB, and ECS**, integrating **TensorFlow / PyTorch**-based chatbot and feedback intelligence, reducing onboarding time by **50%** for **5,000+ MAU** and improving service adoption by **35%**.
- Optimized data access and user workflows with **GraphQL**, reducing data load times by **20%** and improving frontend performance by **30%**, which increased portal usability and engagement.
- Automated deployment and service operations with **Jenkins, GitHub Actions, Docker**, and **Grafana**, increasing release frequency by **40%** and maintaining **99.9%** uptime in production.

Software Engineer, Pantech Prolabs, Pune, India

Jun 2020 - Feb 2021

Distributed Job Platform Data Processing (Primary Skill: Python, Kafka, Apache Spark, Django, Flask)

- Optimized asynchronous data pipelines for a job listing platform using **Python, Kafka, Zookeeper**, and **Apache Spark**, reducing processing time by **30%** and enabling scalable **ETL** workflows for high-volume data handling.
- Enforced secure **RESTful APIs** with **Django, Flask, OAuth2, JWT**, and **Terraform**, protecting data for **100,000+** users, reducing unauthorized access incidents by **45%**, and cutting manual provisioning effort by **35%** through automated, scalable deployment workflows.

PROJECTS

Advanced Lane Detection using Computer Vision and Deep Learning Techniques

Aug 2023 - May 2024

- Devised a lane detection system for autonomous vehicles using **CARLA** simulation, **ENet**, **Hourglass** neural networks, and **RANSAC** algorithms, achieving **93.19%** classification accuracy and **3.89%** MAE, ensuring reliable detection under varied conditions.

Smart Image Store

Sept 2023 - Dec 2023

- Created an Image store where users can upload and retrieve images based on object tags extracted using **Google Cloud Vision** API, optimized search with a HashMap-based algorithm, and utilized **MongoDB**, **React.js**, **load balancing**, and auto-scaling in **AWS**.

EDUCATION

San Jose State University, San Jose, USA

Masters of Science in Software Engineering

GPA: 3.81

Aug 2022 - May 2024

Savitribai Phule Pune University, Pune, India

Bachelors of Engineering

GPA: 3.81

Aug 2016 - May 2020